



UNIVERSITY OF SANTO TOMAS
COLLEGE OF INFORMATION AND COMPUTING SCIENCES
DEPARTMENT OF COMPUTER SCIENCE



**Classification of Rice Leaf Blast Disease Using Hybrid VGG16-Convolutional
Neural Network with Partial Least Squares Regression**

A Thesis

Presented to the

Department of Computer Science

College of Information and Computing Sciences

University of Santo Tomas

In Partial Fulfilment
of the Requirements for the Degree
Bachelor of Science in Computer Science

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November 2025

Research Description

Rice is stipulated as a part of daily meals and one of the primary calorie sources. The Philippines is the sixth-highest consumer at 16.5 million metric tons annually (Leyco C., 2024). However, rice farmers frequently battle diseases such as leaf blast, a destructive fungal pathogen that can severely damage crops and lead to significant losses or even plant death. In severe cases due to severe infection, the crops become useless due to extensive damage to the leaves, resulting in the rice crop's death (Mapuranga et al., 2022).

CNN has been applied for the early detection of crop diseases since the advancement of technology in the agricultural field, such as the study by (Shrivastava et. al., 2021) where VGG16 was used for rice disease detection, achieving an accuracy of 93.11%. Additionally, data augmentation was suggested to increase the dataset artificially. Furthermore, the ability of PLSR to address multicollinearity and dimensionality reduction (Mirandilla et al., 2023), along with data augmentation, is considered by the researchers to be implemented on the system architecture. The researchers have proposed to modify the dense layers of the VGG16-CNN, using the Partial Least Squares Regression (PLSR) model as the classification method to detect the rice leaf blast disease, with the unmodified VGG16-CNN as the baseline and the VGG16-XGBoost model, which was also used for comparison on the performance of the proposed model.

The images for input will be extracted and pre-processed by removing the background and using double-inverse-binary thresholding. Then, the number of images between classes of the foreign dataset was balanced by removing images manually from the higher numbered class. A run with data augmentation was also applied on the training set to enhance the reliability of the model. After pre-processing is completed, the VGG16-CNN model will extract features from the images of the rice leaf blast. Then, three different methods were used to classify the rice leaf using a dense VGG16, PLSR, and XGBoost layers. Lastly, Grad-CAM and SHAP Beeswarm were used to explain the features and better visualize how the classification of the rice leaf blast disease was determined. These models' competitiveness was assessed by training, validation, and testing accuracy. Other performance metrics, such as inference time, precision, recall, F1-score, and specificity, were recorded to compare these three models.

The combination of VGG16-CNN with Partial Least Squares Regression will aid in the early detection of rice leaf blast, with the use of VGG16's ability to extract high-level spatial features from rice leaf images, while using PLSR or XGBoost to enhance classification performance potentially. The research aims to develop a reliable model that can accurately classify rice leaf images despite the variations in the image quality, and assist agricultural professionals, specifically farmers, in identifying rice leaf blast disease from images.